

# SHUVAM PAL

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## Education

**Dayananda Sagar College of Engineering**

Bengaluru, India

*B.E. Artificial Intelligence and Machine Learning*

## Experience

**Bentham AI** 

Dec 2025 – Feb 2026

*Backend Engineering Intern: Go, Node.js, Docker, GCP, Puppeteer, GitHub Actions*

Remote

- Built a **Puppeteer/Node.js automation engine** for stateful MCA filing workflows, reducing manual filing time by **85%** through retries, session recovery, and API-backed state checks.
- Optimized an **AI-assisted company-registration pipeline** for unique company-name suggestions, descriptions, and filing-field drafts, reducing manual drafting/rework across MCA workflow preparation.
- Containerized services with **Docker**, deployed to **Google Cloud Run**, and automated builds/releases with **GitHub Actions**, improving deployment velocity by **70%**.

**Google Summer of Code** 

May 2025 – Sept 2025

*Mentee at Invesalious: Python, VTK, wxPython, NumPy, PyTorch, ONNX*

Remote

- Built MRI segmentation tooling over large volume data, producing masks for **95 brain subparts** with PyTorch/TinyGrad model support.
- Reduced processing runtime by **87%** through async/parallel execution and fewer repeated passes across large image arrays.
- Improved 3D reconstructed surface visualization, keeping derived model outputs inspectable for downstream workflows.

## Open Source

[Contributions](#) 

- Contributed to Invesalious, IOOS, VideoLAN, CRIU, Tiled, Kyverno, and Kuadrant, working with established codebases and maintainer feedback loops.

## Projects

**Seaweed: Contest Platform** | *Go, PostgreSQL, Redis, Judge0, AWS, Docker, Kubernetes*

[GitHub](#) 

- Built a backend-heavy ranking system for **500+ concurrent users**, using PostgreSQL materialized views and concurrent refreshes to preserve live leaderboard behavior.
- Deployed Dockerized services on Kubernetes/AWS and replaced a Redis + DynamoDB design with PostgreSQL-centered architecture, reducing infrastructure cost by **70%**.

**Distributed Inference Engine** | *Python, PyTorch, FastAPI, gRPC, CUDA, Docker, Redis*

- Built a **distributed inference engine** to run models larger than a single consumer GPU by coordinating token generation across heterogeneous nodes with speculative decoding.
- Designed streaming APIs, worker heartbeats, retry-aware orchestration, and KV-cache-aware routing to balance latency, throughput, and worker availability across local and remote inference workers.

**Postificus** | *Go, Echo, Rod, PostgreSQL, Redis, RabbitMQ, Docker, AWS, Prometheus*

[GitHub](#) 

- Built observable Go services for a **content distribution platform** using browser automation, with RabbitMQ workers, DLQs, retries, Redis autosave, and PostgreSQL persistence.
- Added circuit-breaker logic and failure isolation for third-party publishing paths, keeping browser automation failures observable and recoverable instead of cascading through the system.

## Technical Skills

**Languages:** Go, Python, TypeScript, C++

**Frameworks:** FastAPI, Echo, Node.js, gRPC

**Tools:** PostgreSQL, Redis, RabbitMQ, Docker, Prometheus, AWS, GitHub Actions